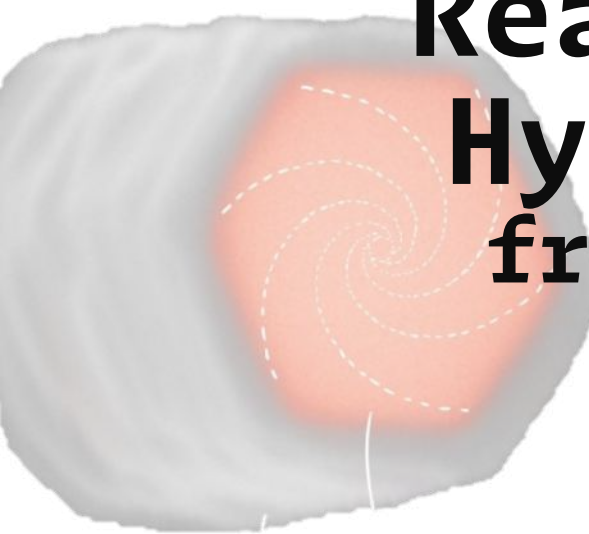




Realistic Modeling of 1D Hybrid Superconductors: from equations to experimental simulations

Carlos Payá

International Workshop on Superconductor-Semiconductor Hybrids
Villard-de-Lans, France – 5 February 2026



CSIC

icmm

Our team



Elsa Prada
Thesis supervisors

Ramón Aguado

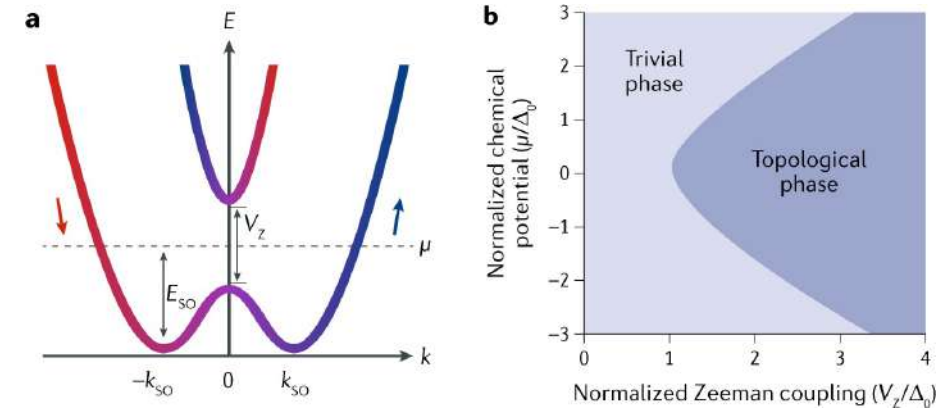
Pablo San-Jose
Quantica.jl

icmm

Toy models are not enough



$$H = \frac{p_x^2}{2m} - \mu + V_Z + \alpha p_x \sigma_y + \Delta \tau_x$$



R. M. Lutchyn et al. *Nature Reviews Materials* 3, 52-68 (2018)

But actual devices...

Disorder

Inhomogeneities

3D (multiband)

1) Full *ab initio* models

2) Elaborated models, close to the experiment but in touch with theory

A multi-level pipeline



Full-shell Hybrid Nanowires

Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

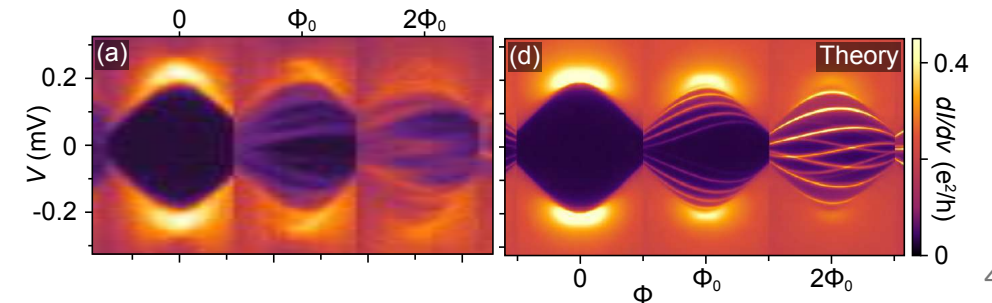
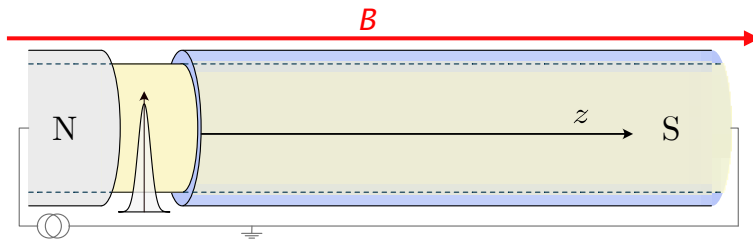
P. San-Jose, CP, C. M. Marcus, S. Vaitiekėnas, E. Prada
Phys. Rev. B 107, 155423 (2023)

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

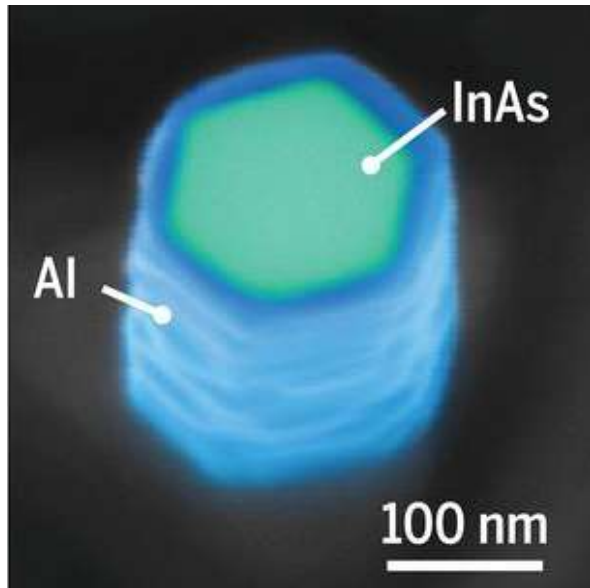
CP, S. D. Escribano, A. Vezzosi, F. Peñaranda, R. Aguado, P. San-Jose, E. Prada
Phys. Rev. B 109, 155428 (2024)

Caroli-de Gennes-Matricon Analogs in Full-Shell Hybrid Nanowires

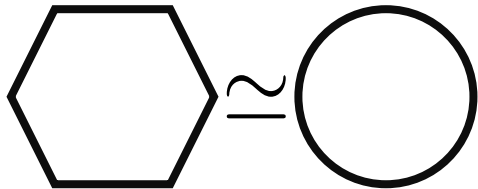
M. T. Deng, CP, P. San-Jose, E. Prada, C. M. Marcus, S. Vaitiekėnas
Phys. Rev. Lett. 134, 206302 (2025)



Modelling the device

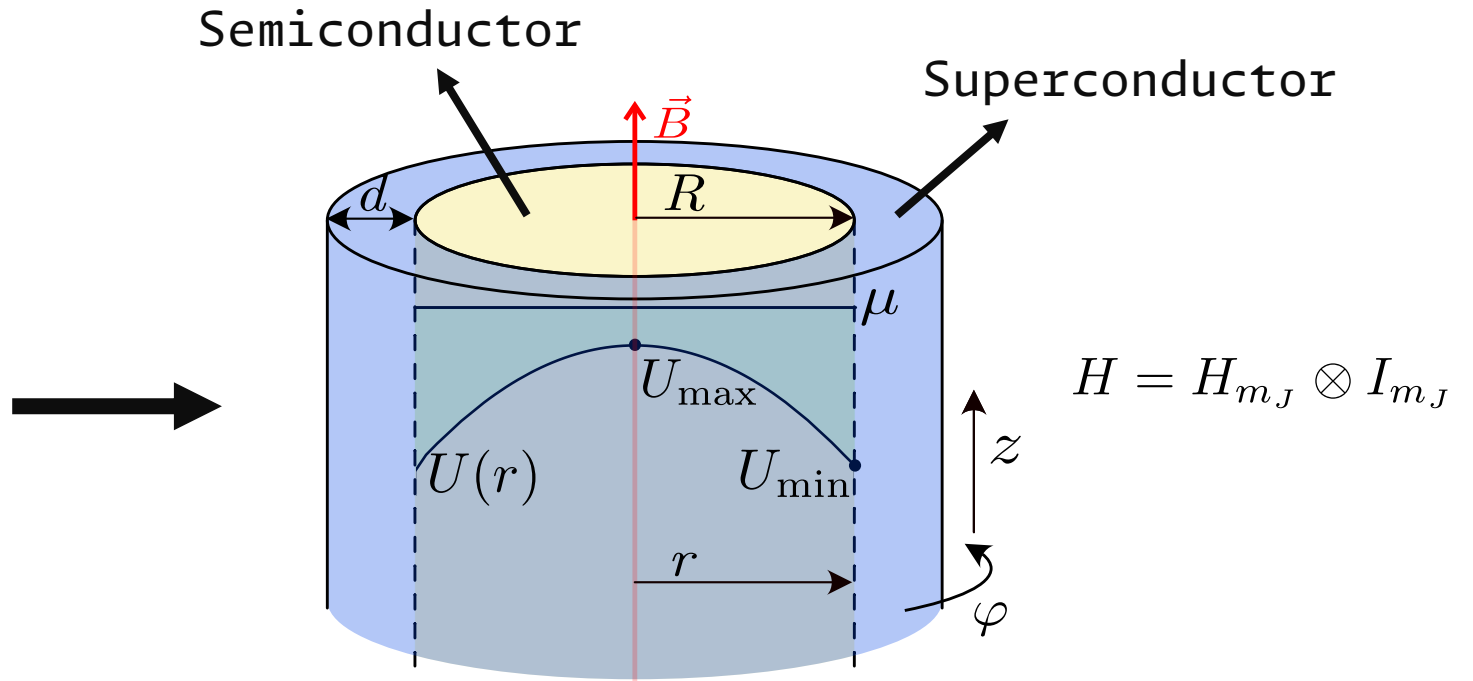


S. Vaitiekėnas et al. *Science* 367, 1442 (2020)



Semiconductor

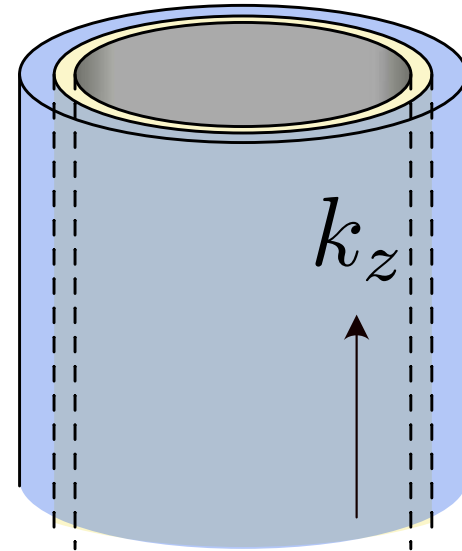
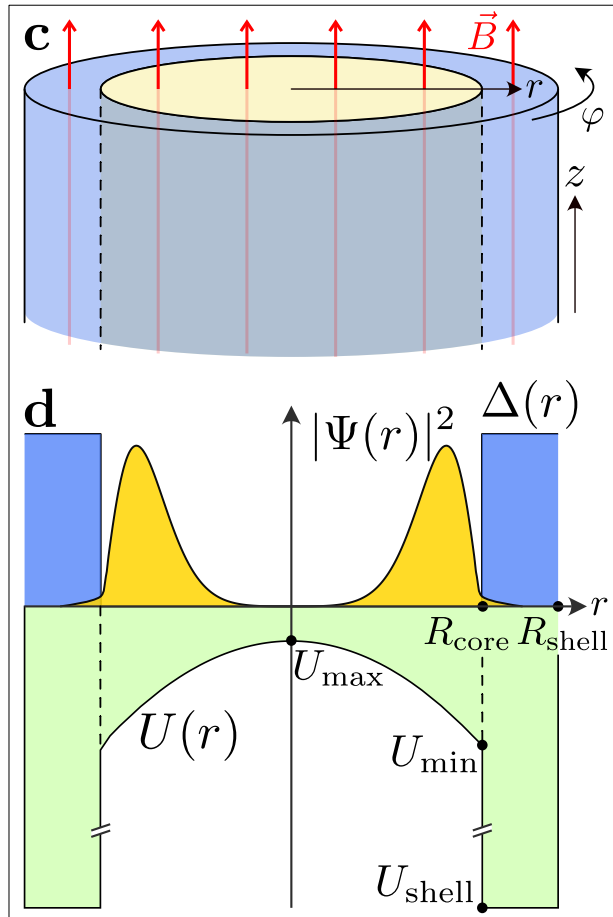
Superconductor



$$H = H_{m_J} \otimes I_{m_J}$$

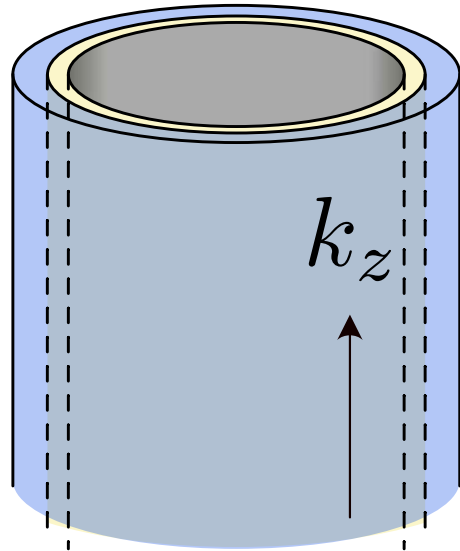
- Cylindrical symmetry
- Charge at interface
- SOC induced by band bending

Modelling the device



$$r \rightarrow R$$

Modelling the device



Rashba wire

$$H_{\text{SM}} = \frac{(\vec{p} + eA_\varphi \hat{\varphi})^2}{2m^*} - \mu + \alpha \hat{r} \cdot [\vec{\sigma} \times (\vec{p} + eA_\varphi \hat{\varphi})] + V_Z \sigma_z$$

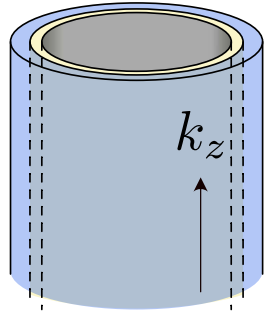
Diffusive superconductor

$$\Sigma_{\text{SC}}(\omega) = \Gamma \frac{\tau_x - u(\omega)\tau_0}{\sqrt{1 - u^2(\omega)}}$$

BdG formalism, Nambu basis $\begin{pmatrix} \Psi_\uparrow, \Psi_\downarrow, \Psi_\downarrow^\dagger, -\Psi_\uparrow^\dagger \end{pmatrix}$
 $\vec{\sigma} \otimes \vec{\tau}$

$$H_{m_J} = \langle m_J | H(\varphi) | m_J \rangle$$

Doubly connected SC: Little-Parks modulation



Modulation of the SC pairing with magnetic field

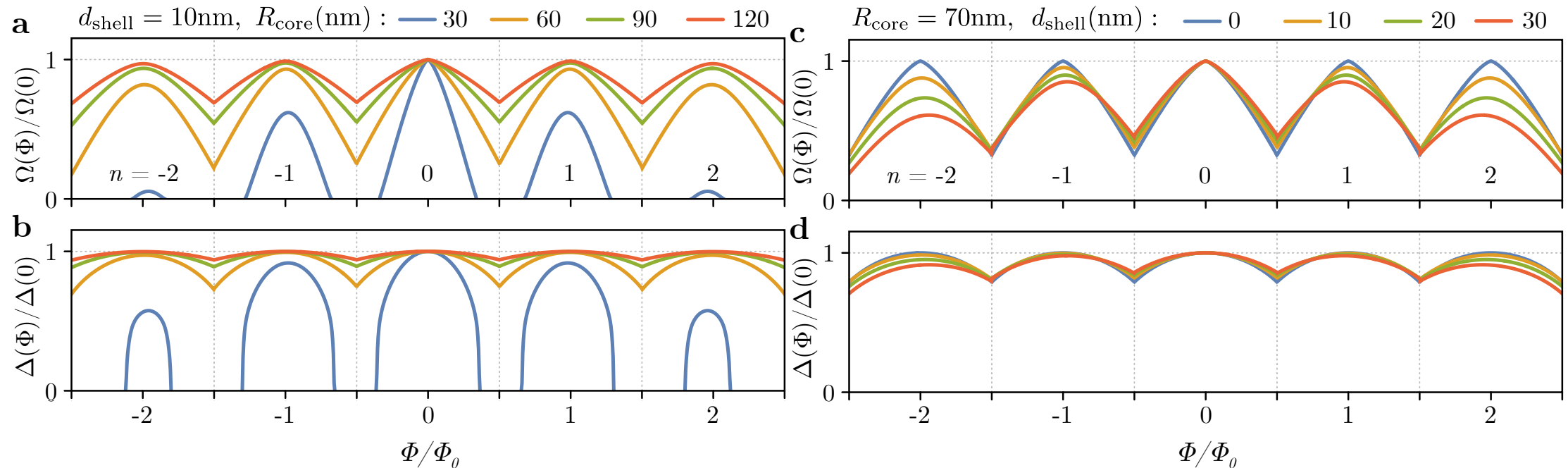
$$\Delta = |\Delta| e^{in\varphi}$$

$$n = \frac{\Phi'}{\Phi_0}, \Phi = \Phi' + \oint \Lambda \vec{J} d\vec{l}$$

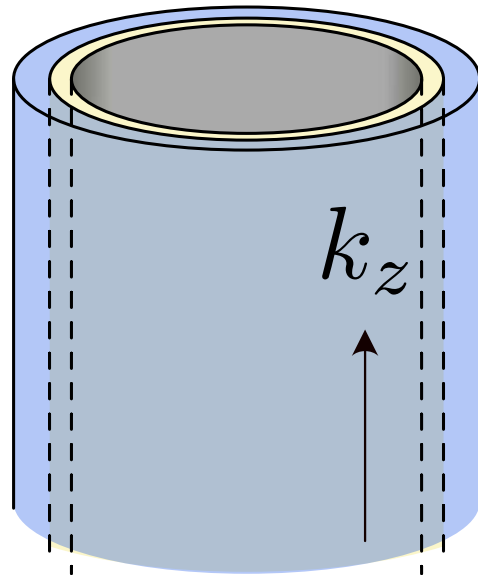


supercurrent

Diffusive SC $\Rightarrow \Delta \neq \Omega$



Doubly connected SC: Little-Parks modulation



$$\Sigma_{\text{SC}}(\omega) = \Gamma \frac{\tau_x - u(\omega)\tau_0}{\sqrt{1 - u^2(\omega)}}$$

Usadel

$$u(\omega) = \frac{\omega}{\Delta(\Lambda)} + \frac{\Lambda}{\Delta(\Lambda)} \frac{u(\omega)}{\sqrt{1 - u^2(\omega)}}$$

Depairing

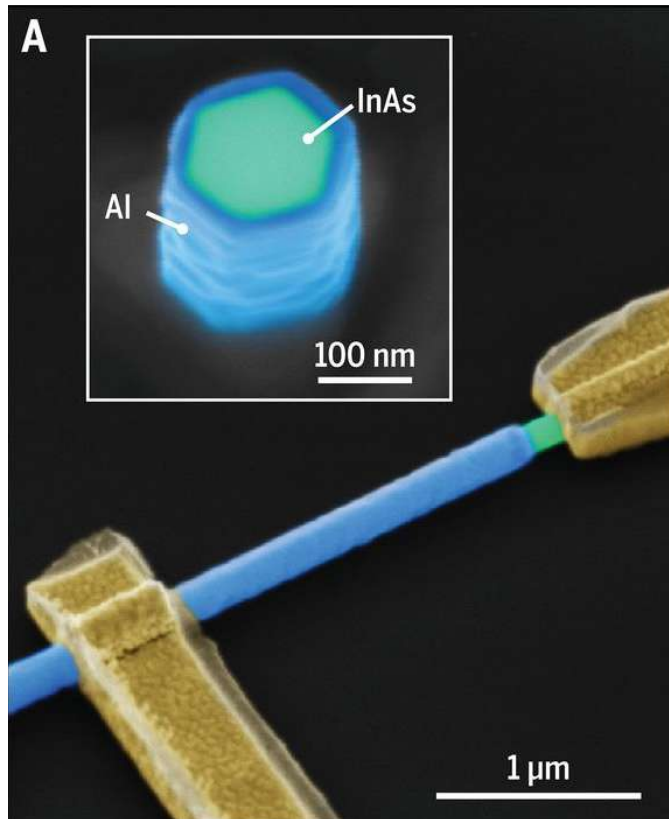
Magnetic flux $\rightarrow$

$$\Lambda(\Phi) \propto \frac{\xi_d^2}{(R + d/2)^2} \left(n - \frac{\Phi}{\Phi_0} \right)^2 + \frac{d^2}{(R + d/2)^2} \left(\frac{\Phi^2}{\Phi_0^2} + \frac{n^2}{3} \right)$$

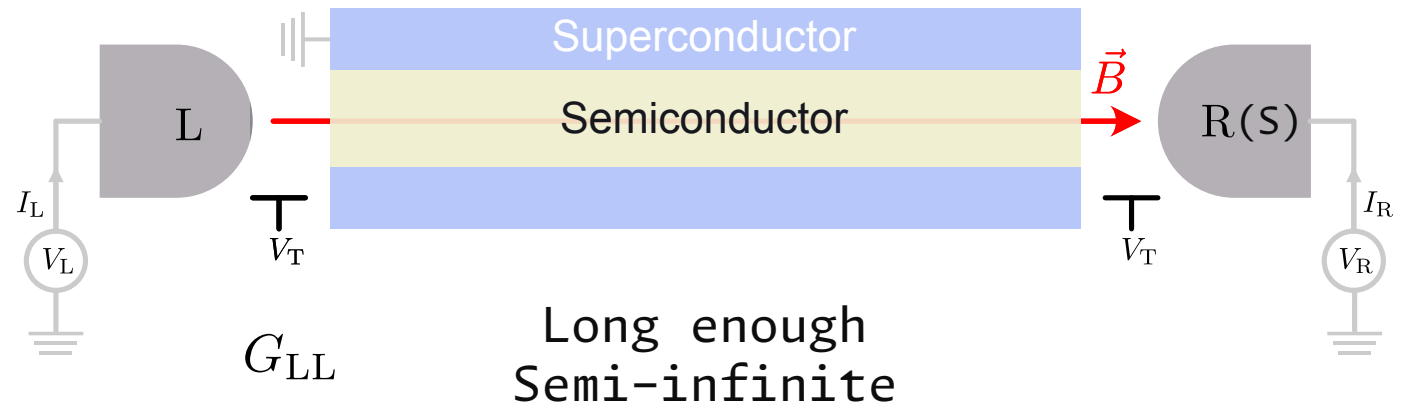
$$n = \left\lfloor \frac{\Phi}{\Phi_0} \right\rfloor$$

Geometry treated as paramagnetic impurities
(Abrikosov-Gor'kov, 1961)

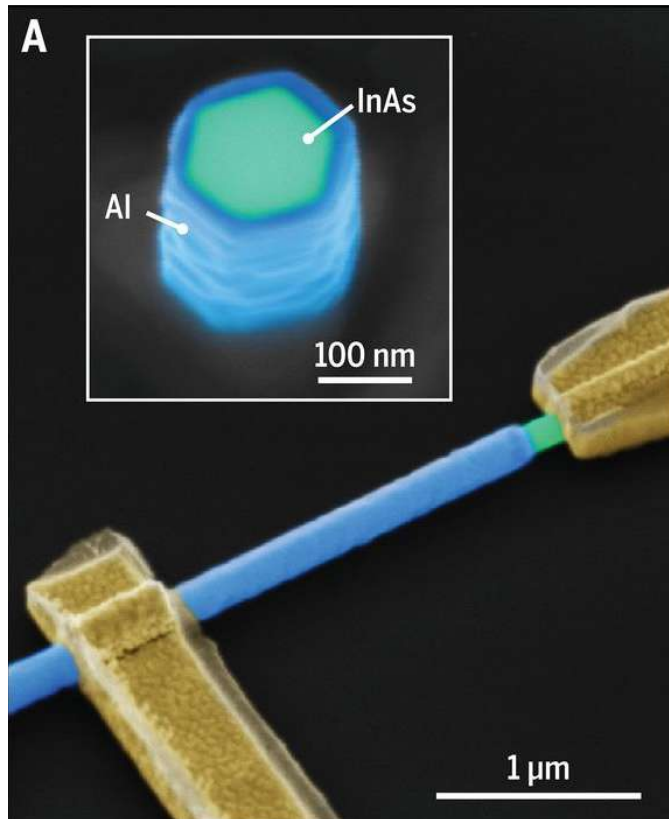
Modelling the setup



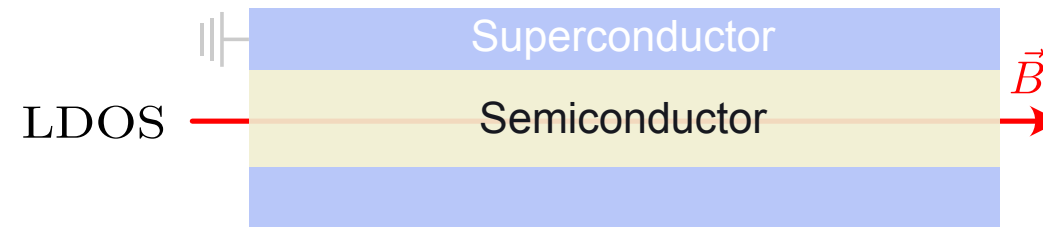
S. Vaitiekėnas *et al.* *Science* 367, 1442 (2020)



Modelling the setup



S. Vaitiekėnas *et al.* *Science* 367, 1442 (2020)

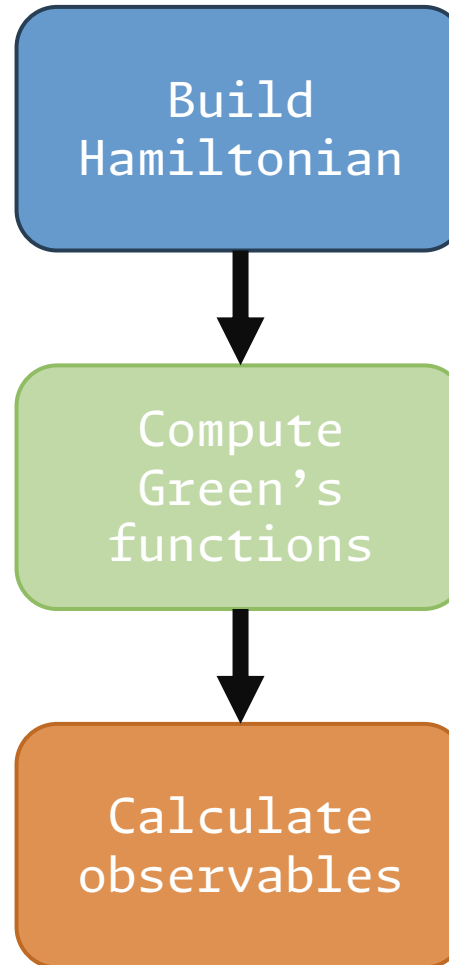


Long enough
Semi-infinite

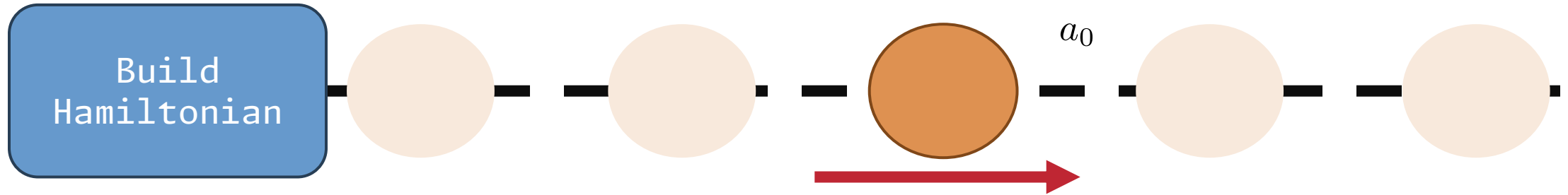
Quantica.jl: our tight-binding toolkit



Pablo San-Jose



Quantica.jl: our tight-binding toolkit



Lattice

+

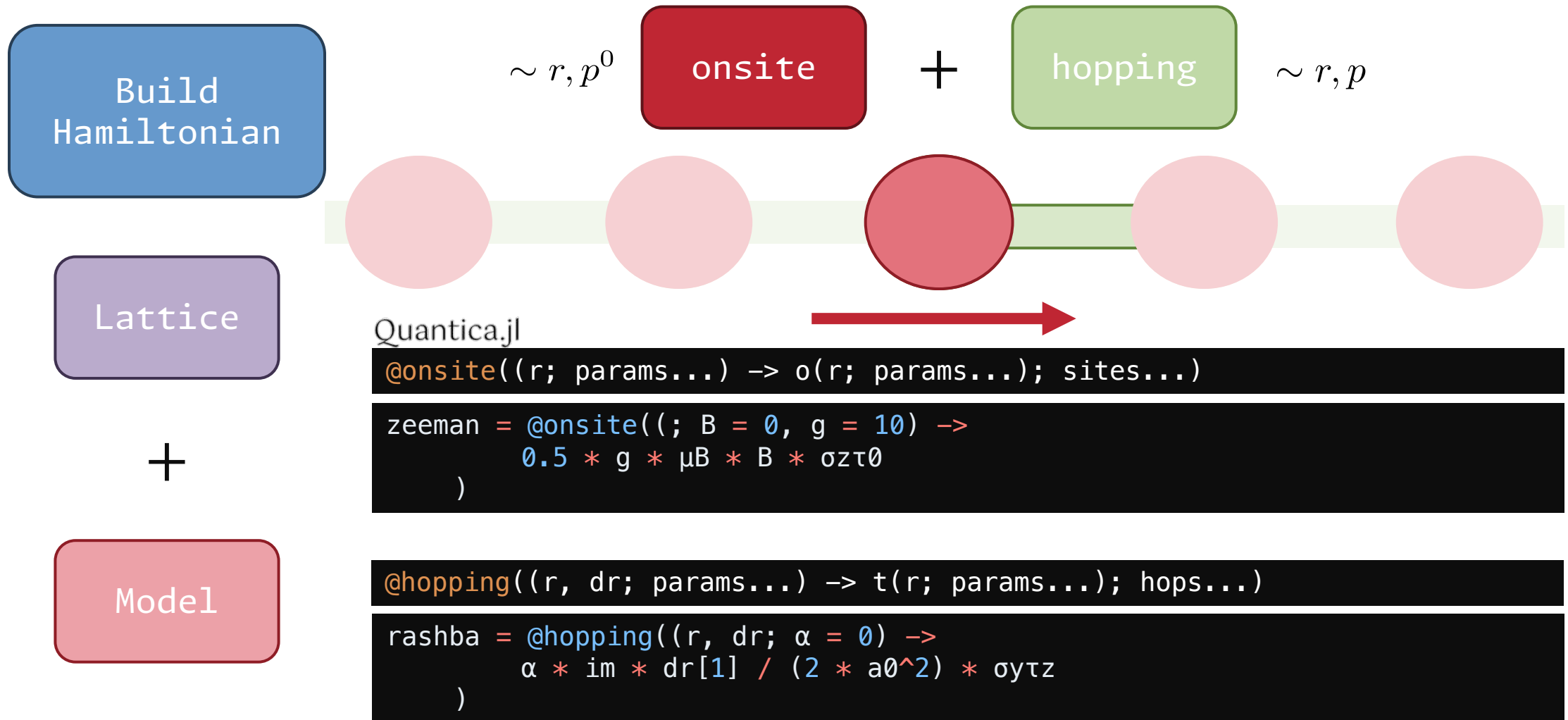
Model

Quantica.jl

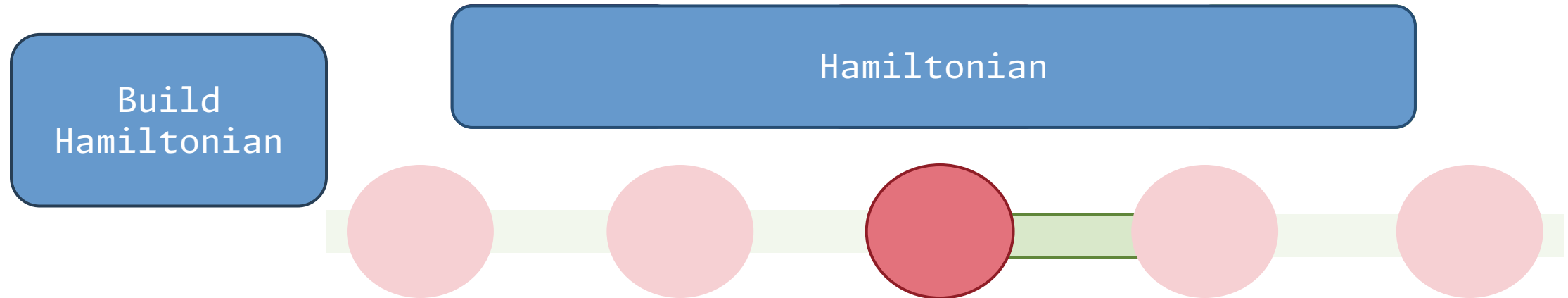
```
lat = LatticePresets.linear(; a0 = 5)
```

```
julia> lat
Lattice{Float64,2,1} : 1D lattice in 1D space
Bravais vectors : [[5.0]]
Sublattices      : 1
Names            : (:A,)
Sites            : (1,) --> 1 total per unit cell
```

Quantica.jl: our tight-binding toolkit



Quantica.jl: our tight-binding toolkit



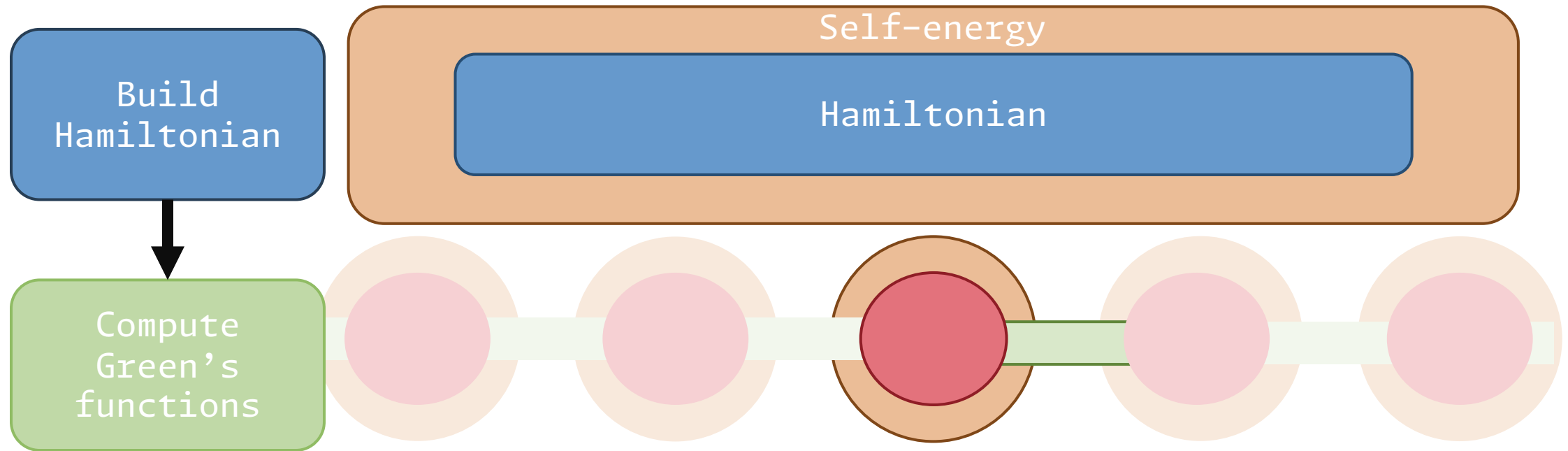
$$H(\phi) = \sum_{dn} H_{dn} e^{-i\phi dn}$$

Quantica.jl

```
hSM = lat |> hamiltonian(kinetic + rashba + zeeman + kinetic_gauge + rashba_gauge; orbitals = 4)
```

```
julia> hSM(π/4; B = 0.3)
4×4 SparseArrays.SparseMatrixCSC{ComplexF64, Int64} with 16 stored entries:
 39.0567+2.58552e-15im    0.0-2.82843im    0.0+0.0im    0.0+0.0im
 0.0+2.82843im    36.5218+2.58552e-15im    0.0+0.0im    0.0+0.0im
 0.0+0.0im    0.0+0.0im    -36.678-2.58552e-15im    0.0+2.82843im
 0.0+0.0im    0.0+0.0im    0.0-2.82843im    -39.5809-2.58552e-15im
```

Quantica.jl: our tight-binding toolkit

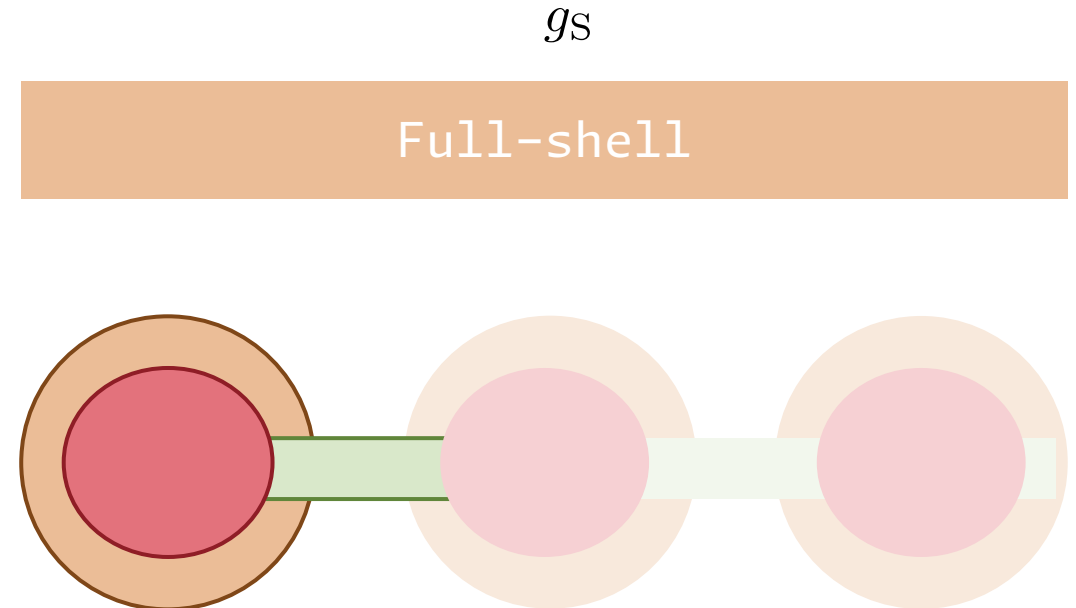
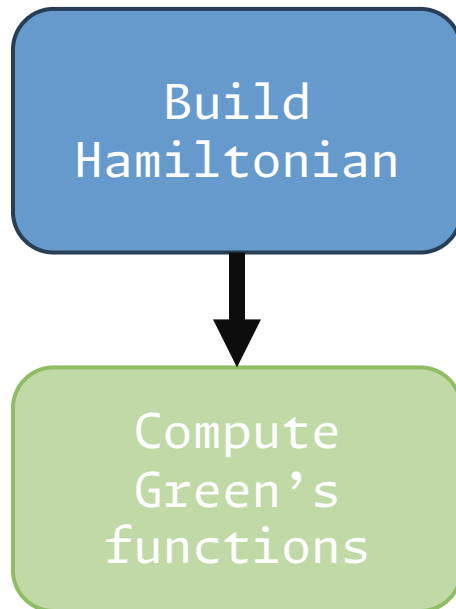


$$G = \frac{1}{\omega - H - \Sigma}$$

Quantica.jl

```
# Quantica modifier to apply self-energy to onsite terms
Σ! = @onsite!((o, r; ω = 0, B = 0, Γ = Δ) ->
    o + Γ * Σ(ω, Λ(B))
)
hSC = hSM |> Σ!
```

Quantica.jl: our tight-binding toolkit



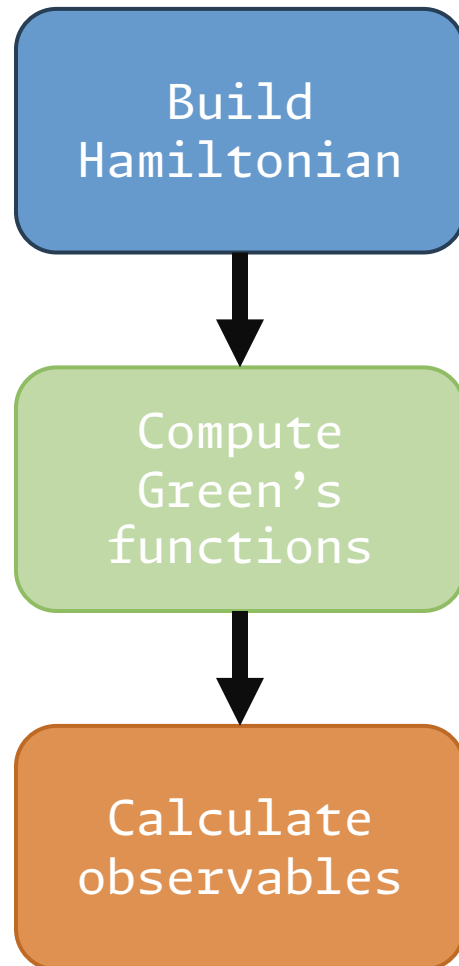
$$G = \frac{1}{\omega - H - \Sigma}$$

Quantica.jl

```
# Quantica modifier to apply self-energy to onsite terms
Σ! = @onsite!((o, r; ω = 0, B = 0, Γ = Δ) ->
              o + Γ * Σ(ω, Λ(B))
              )
hSC = hSM |> Σ!
```

```
gS = hSC |> greenfunction(GreenSolvers.Schur(boundary = 0))
```

Quantica.jl: our tight-binding toolkit



g_S

Full-shell

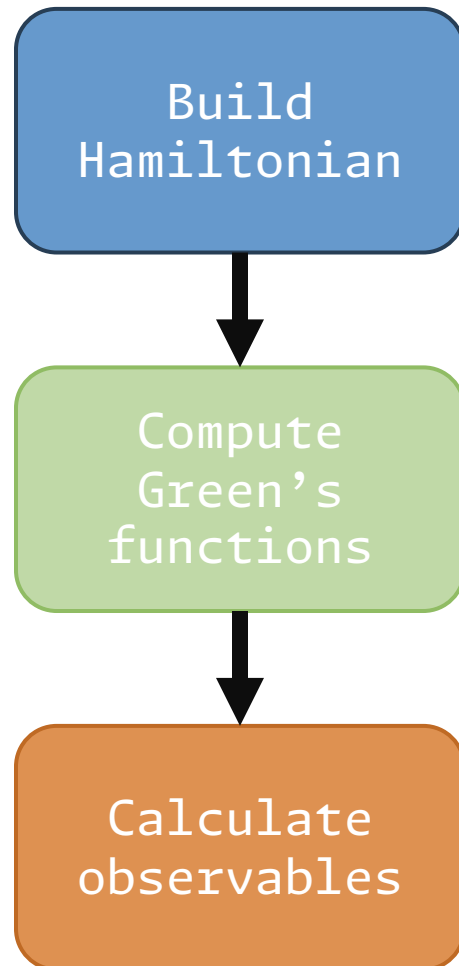
$$\text{LDOS} = -\frac{1}{\pi} \text{Im} \{ \text{Tr} [g_S] \}$$

Quantica.jl

```
julia> ρ = ldos(gS[cells = (1,)]; kernel = I)
LocalSpectralDensitySlice{Float64} : local density of states at a fixed
location and arbitrary energy
  kernel    : LinearAlgebra.UniformScaling{Bool}(true)

julia> ω = 0.1e-3
julia> ρ(ω; ω, B = 0.1)
1-element OrbitalSliceVector{Float64,Vector{Float64}}:
 0.001538907253553478
```

Quantica.jl: our tight-binding toolkit



February 9, 2026

Quantica.jl

```
pts = Iterators.product(E_vals, B_vals, Z_vals)
```

```
LDOS = @showprogress pmap(pts) do (E, B, Z)
    return ρ(E; B = B, ω = E, Z = Z)[1]
end
```

```
carlospaya@cap-mb Tutorial_QuanticaHybrids % julia --project main.jl
Activating project at CENSORED
```

```
=====
RUNNING LDOS CALCULATION
=====
```

```
Parameters:
  Energy points: 301
  Field points: 300
  Z sectors: 11 (from -5 to 5)
```

```
Total evaluations: 993300
```

```
=====
Running parameter sweep...
```

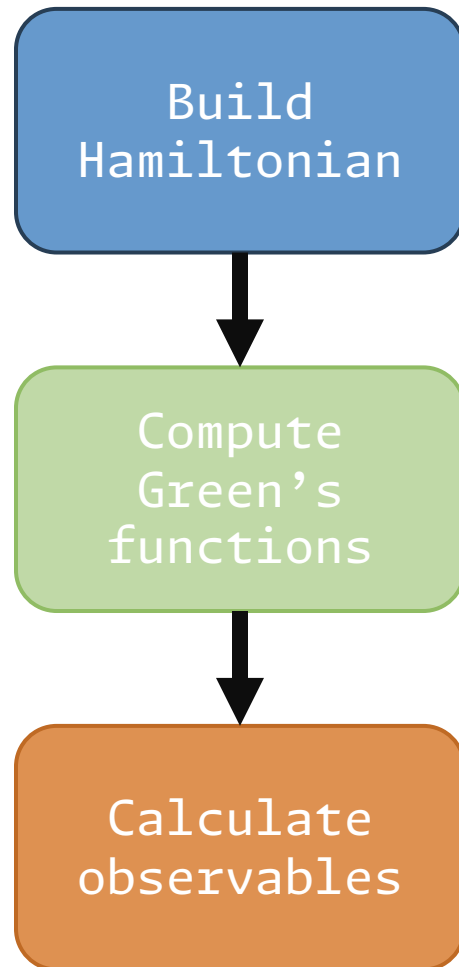
```
Progress:
```

```
100%|
```

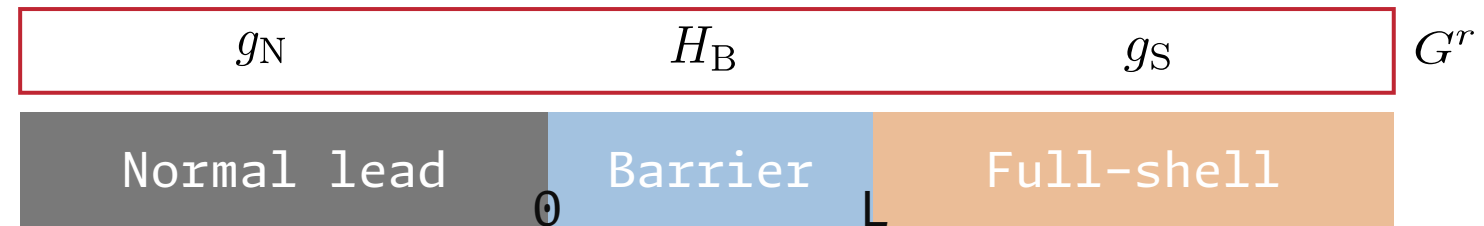
```
| Time: 0:02:26
```

```
151.871329 seconds (195.82 M allocations: 15.532 GiB, 1.41% gc time, 506 lock
conflicts, 4.65% compilation time)
```

Quantica.jl: our tight-binding toolkit



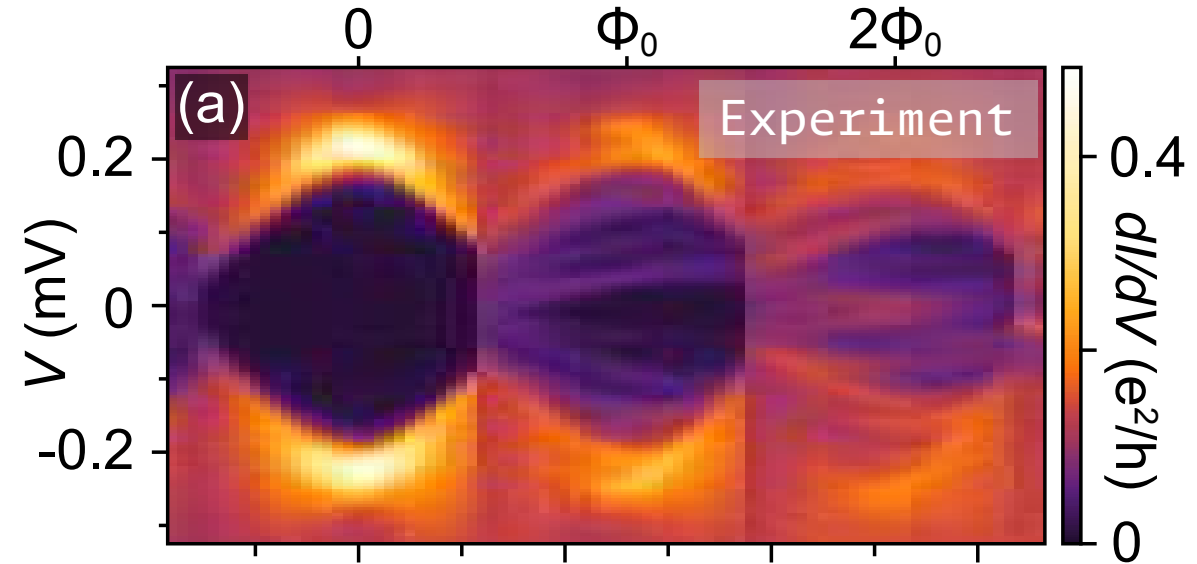
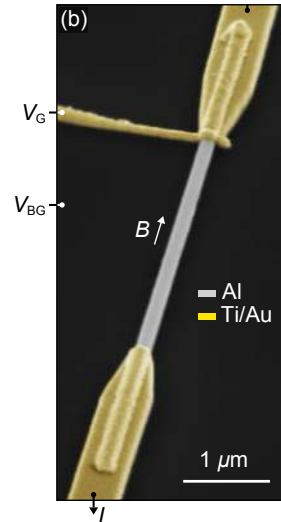
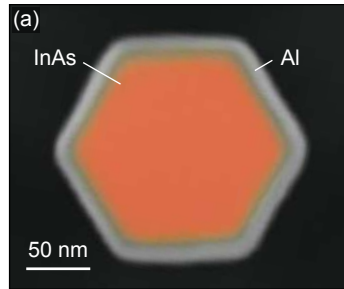
For Conductance: attach a barrier and a lead



Quantica.jl

```
Gr = hB |> attach(gN; region = r -> r[1] == 0) |> attach(gS; region = r ->  
r[1] == L) |> greenfunction()  
G = conductance(Gr[1, 1]; nambu = true)
```

CdGM: experiment vs calculation



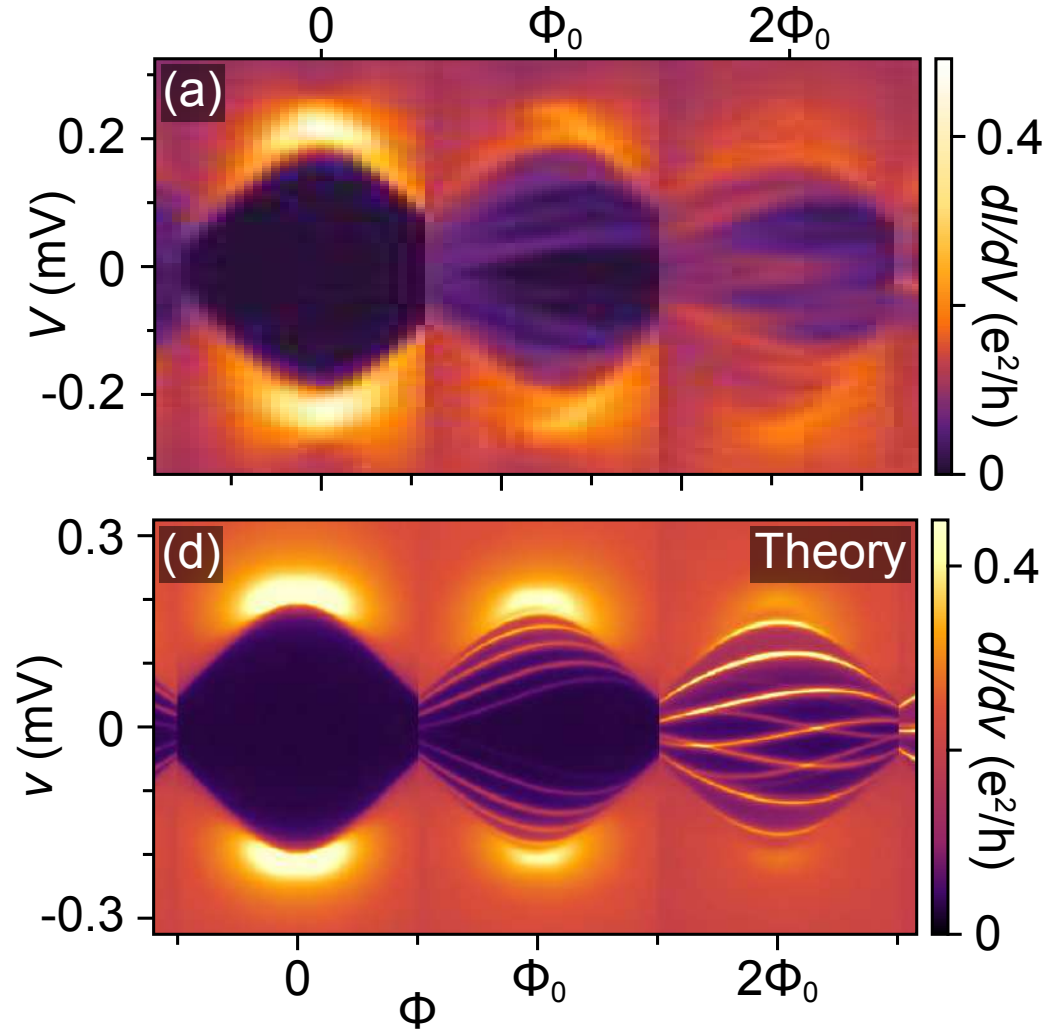
Number of CdGMs $\rightarrow$ Chemical potential: $\mu_{m_L} = \frac{1}{2m^*} \left(\frac{m_L^2}{R_{av}^2} + \frac{1}{4R_{av}^2} + \frac{1}{4} \frac{R_{av}^2}{R_{LP}^4} - \frac{1}{2R_{LP}^2} \right)$

Effective gap $\rightarrow$ Coupling to superconductor: $\Gamma = \Delta_0^* \sqrt{\frac{\Delta_0 + \Delta_0^*}{\Delta_0 - \Delta_0^*}}$

$$\omega_d = \epsilon_0(m_L) - \epsilon_0(m_L - 1) = \frac{1}{2m^*} \left(\frac{1}{R_{av}^2} - \frac{1}{R_{LP}^2} \right)$$

Low-energy CdGM spacing $\rightarrow$ Radial distribution: $\omega_{m_L} \simeq \Delta_0 \left(\frac{\Gamma^2 - (m_L \omega_d)^2}{\Gamma^2 + (m_L \omega_d)^2} \right)$

CdGM: experiment vs calculation



Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

M. T. Deng, CP, P. San-Jose, E. Prada,

C. M. Marcus and S. Vaitiekėnas

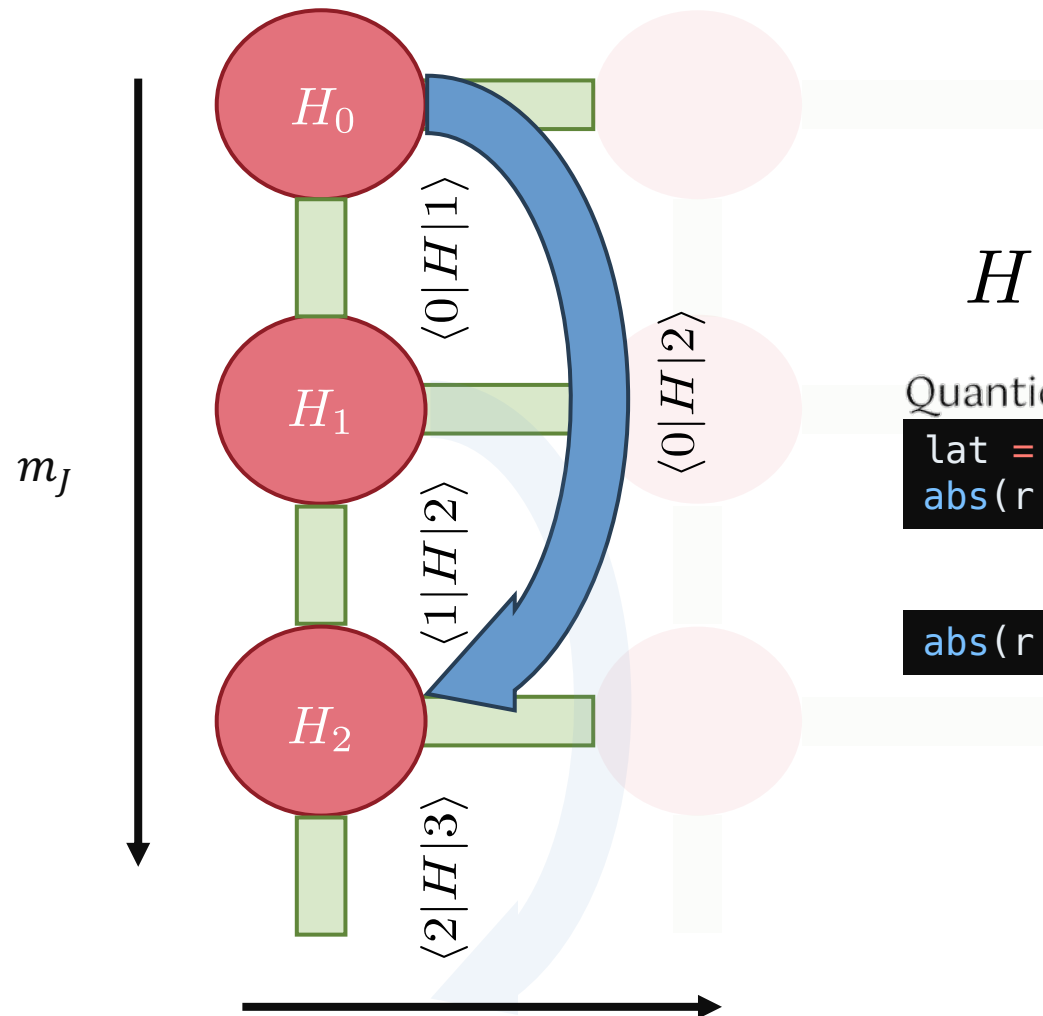
Phys. Rev. Lett. **134**, 206302 (2025)



UNIVERSITY OF
COPENHAGEN



Expanding the model: multi-mode



$$H = \sum_{m_J} H_{m_J} = \text{Tr} [H_{m_J} \otimes I_{m_J}]$$

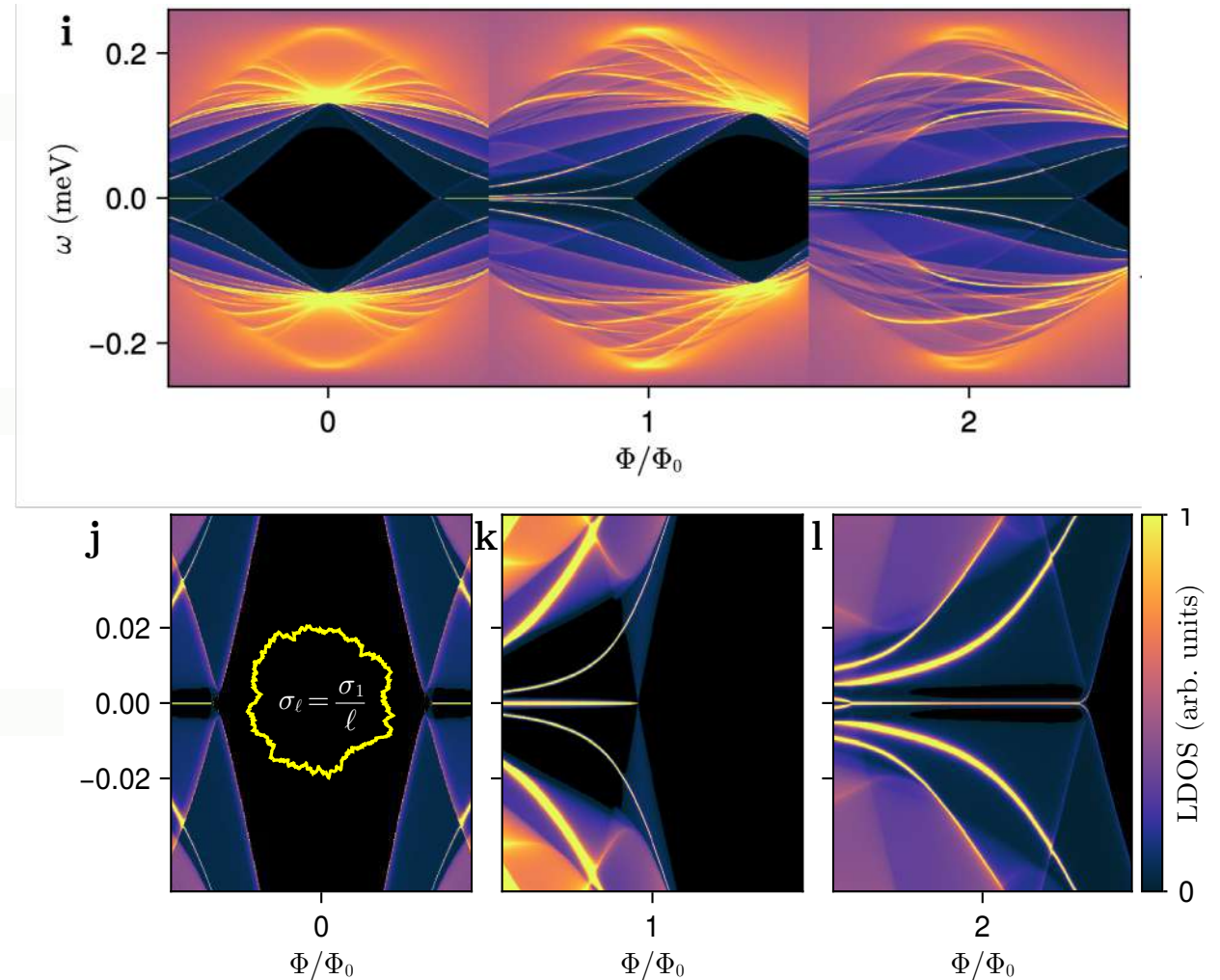
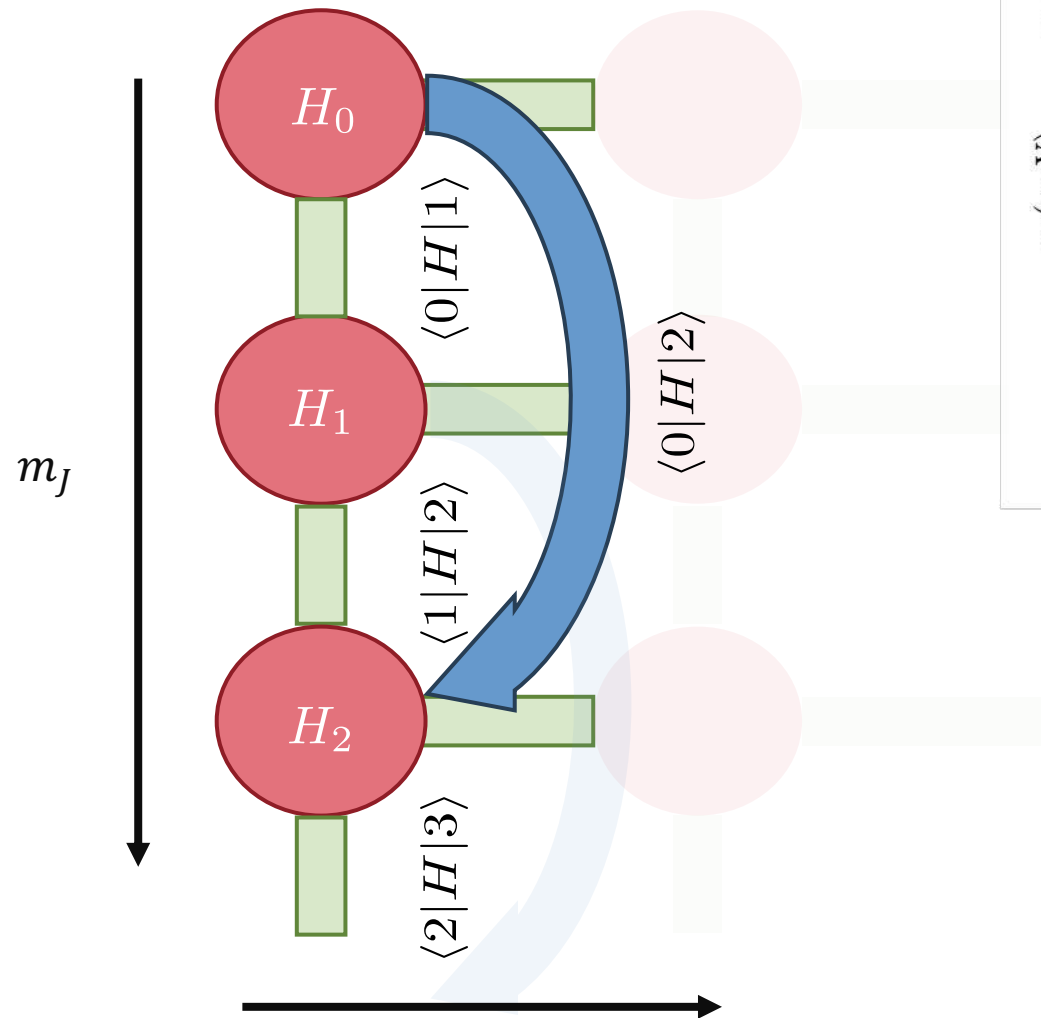
$$H = H_{m_J} \otimes I_{m_J} + \sum_{\ell} \langle m_J | H | m_J + \ell \rangle$$

Quantica.jl

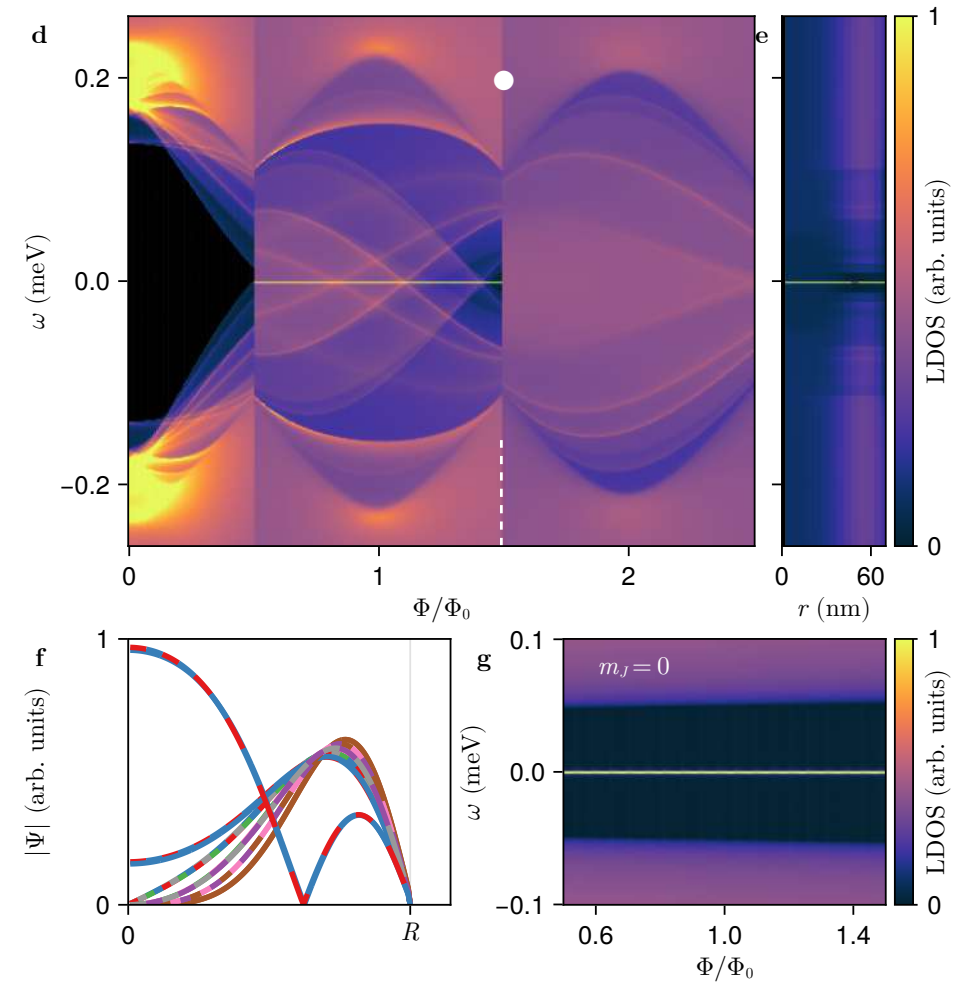
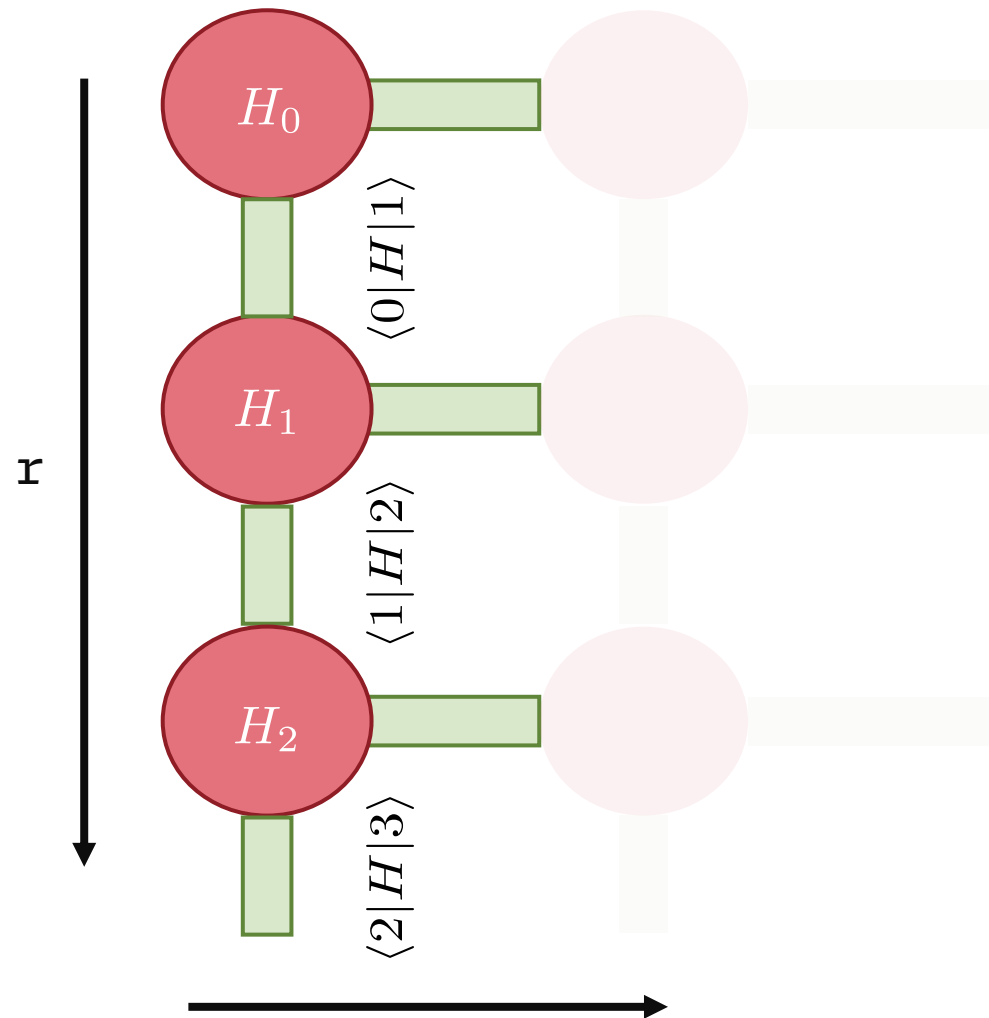
```
lat = LP.square(; a0) |> supercell((1,0), region = r ->
abs(r[2]/a0) <= num_mJ)
```

`abs(r[2]/a0)` is m_J

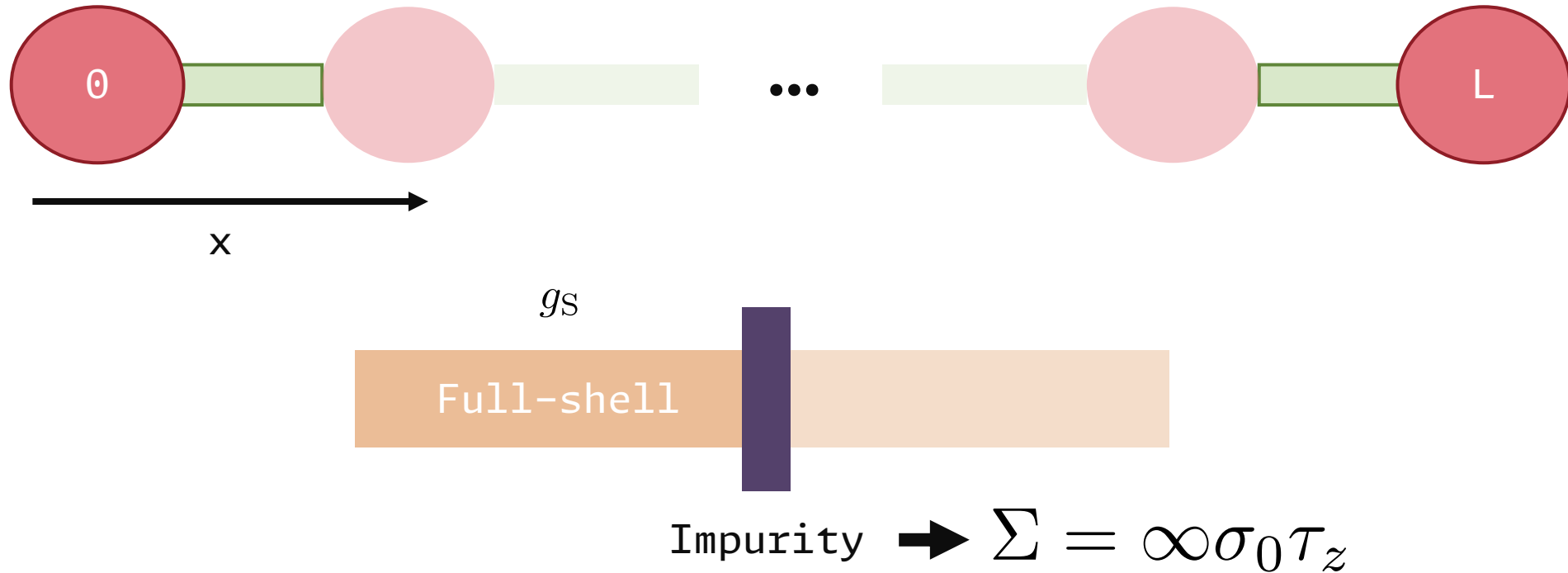
Expanding the model: multi-mode



Expanding the model: multi-mode



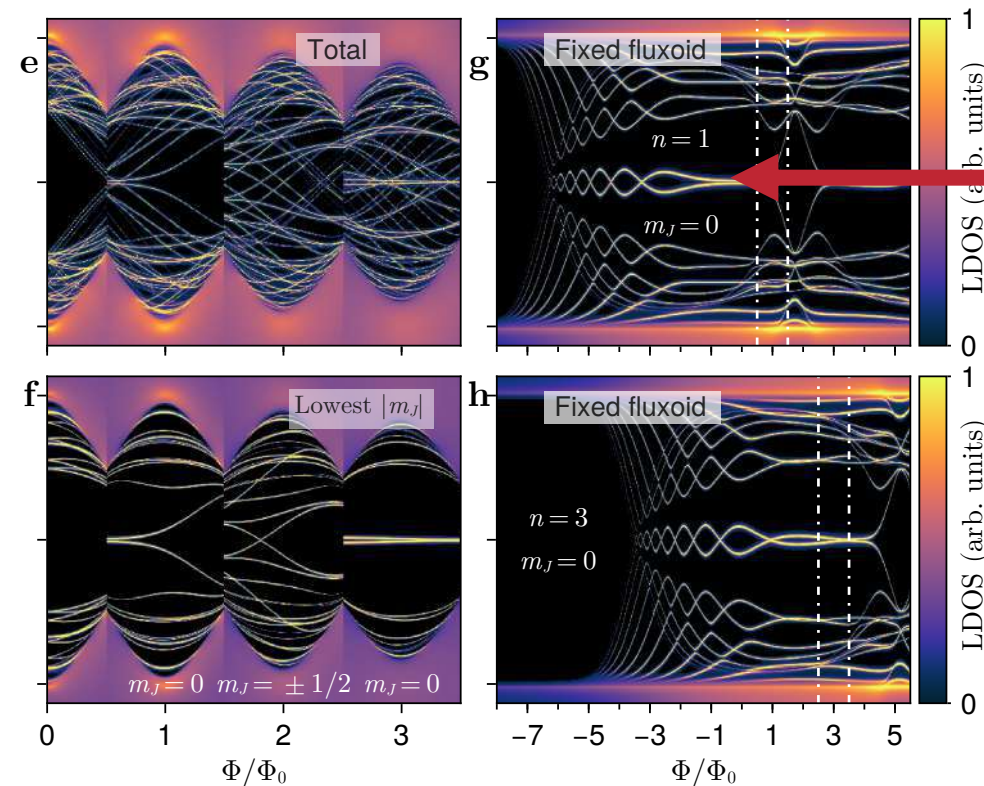
Expanding the model: finite length



Quantica.jl

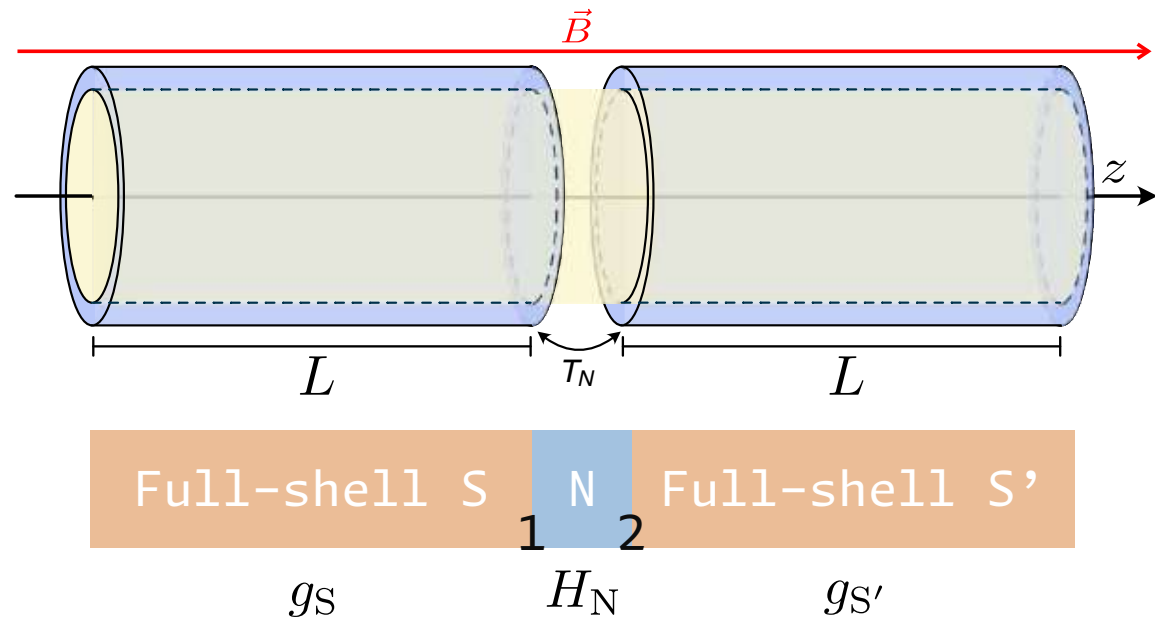
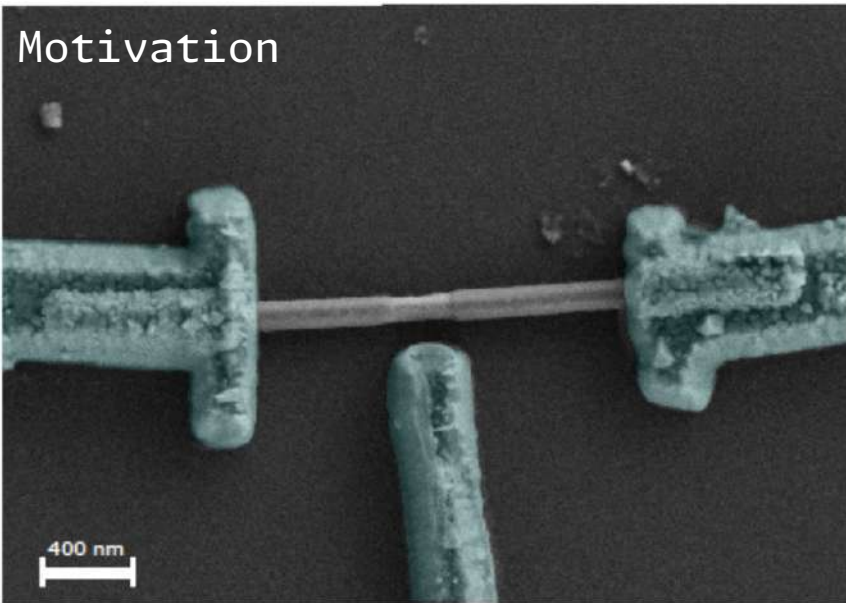
```
g = hSC |> attach(onsite(1e9 *  $\sigma_0 \tau_z$ ), cells = (L,)) |> greenfunction(GS.Schur(boundary = 0))
```

Expanding the model: finite length



Majorana oscillations are larger than a Little-Parks lobe

Building other setups: Josephson Junction

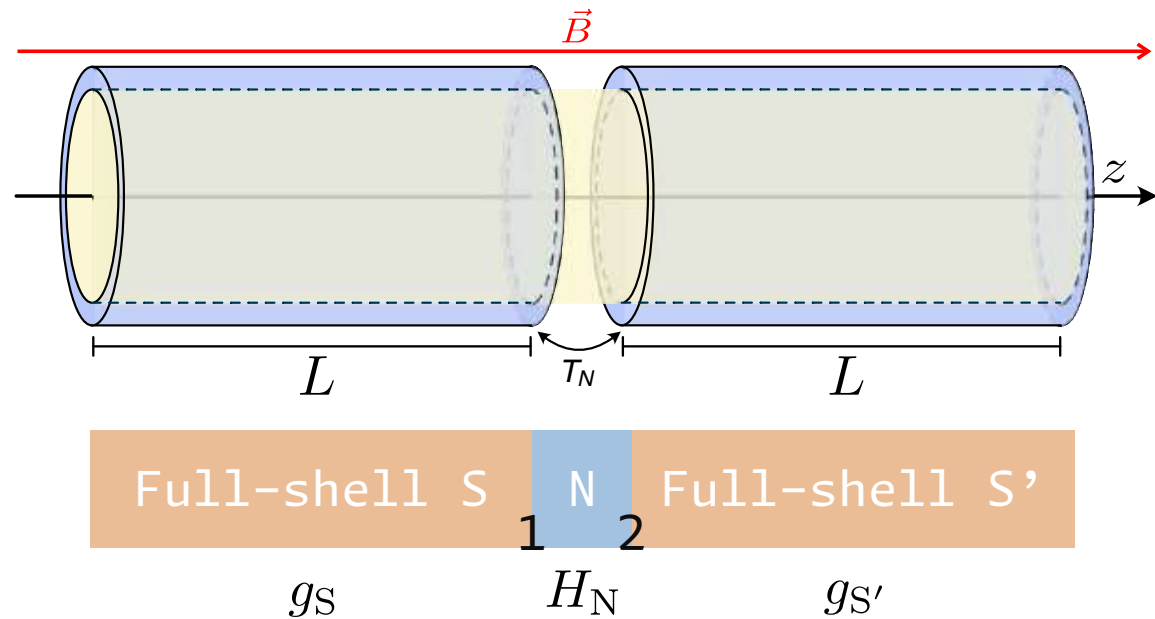
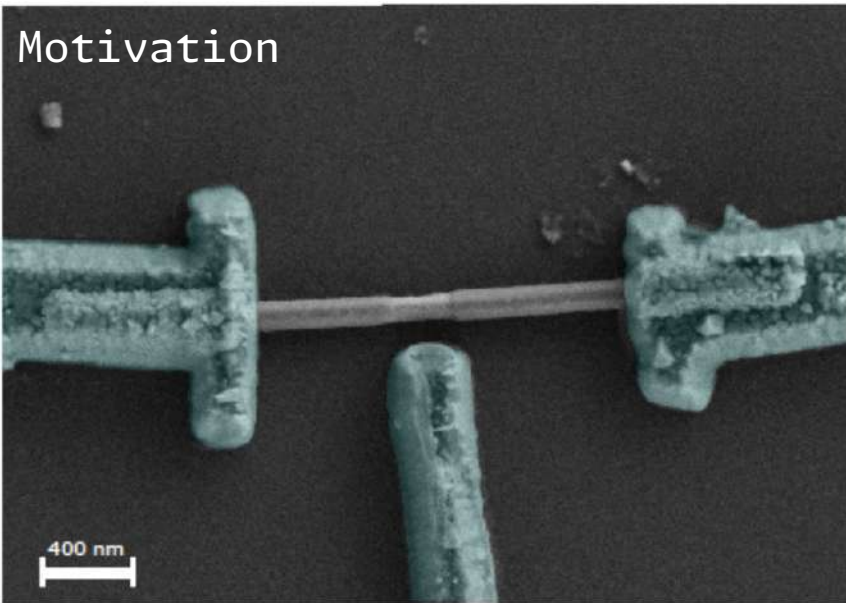


Quantica.jp

Eduardo Lee



Building other setups: Josephson Junction



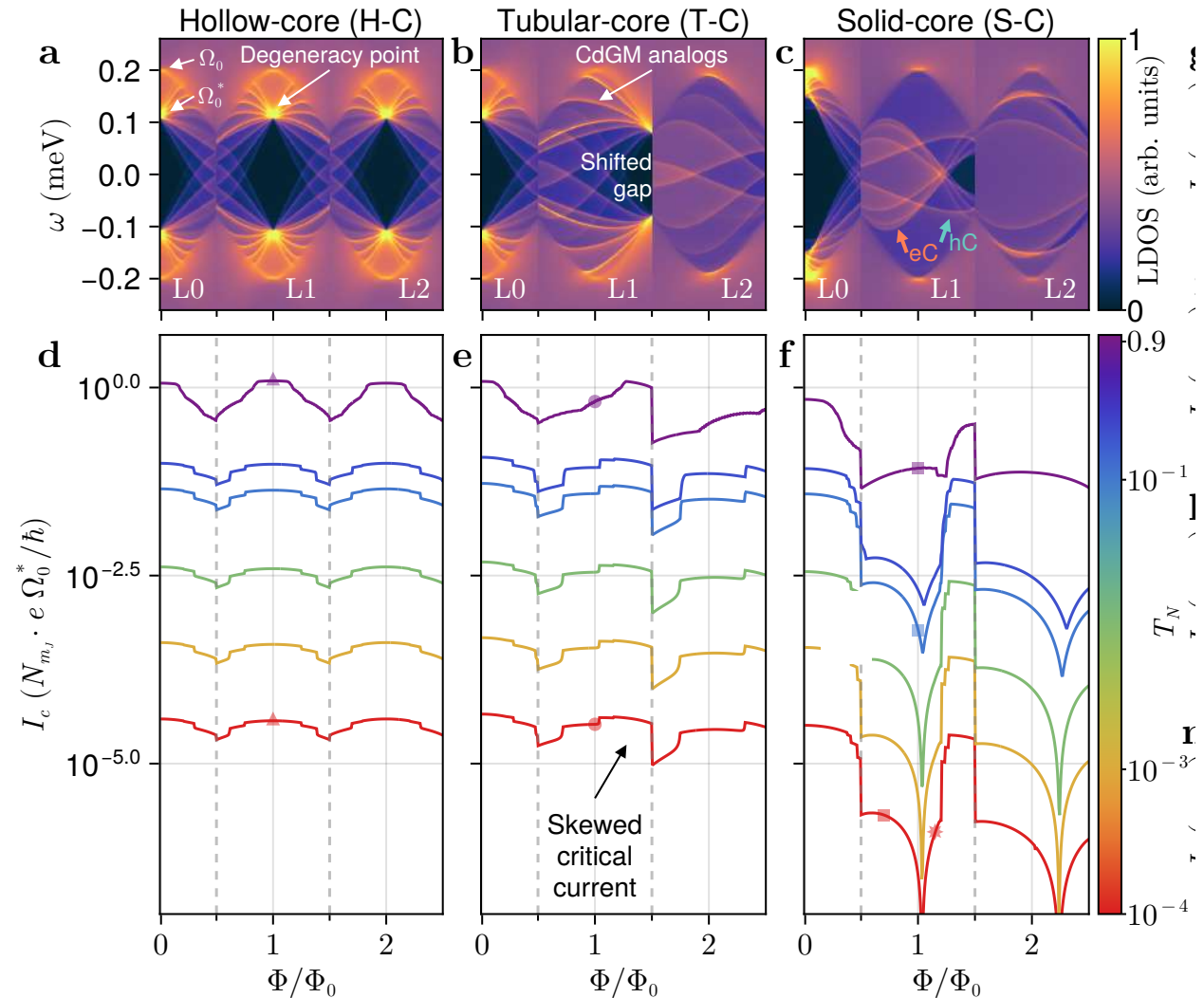
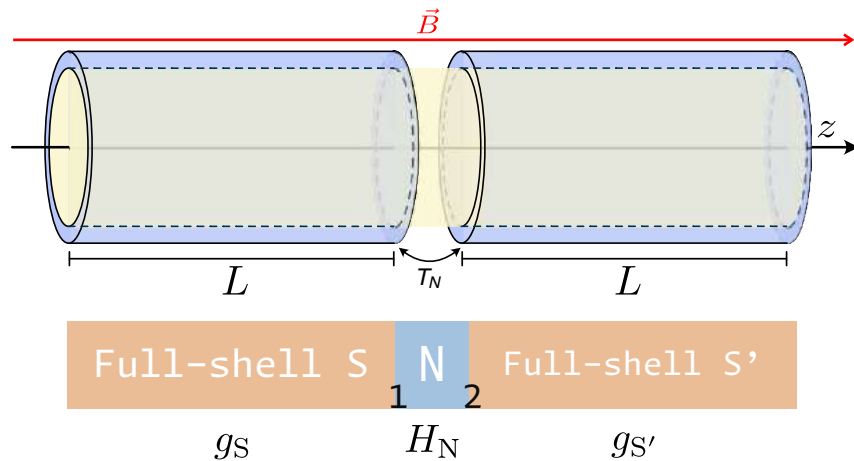
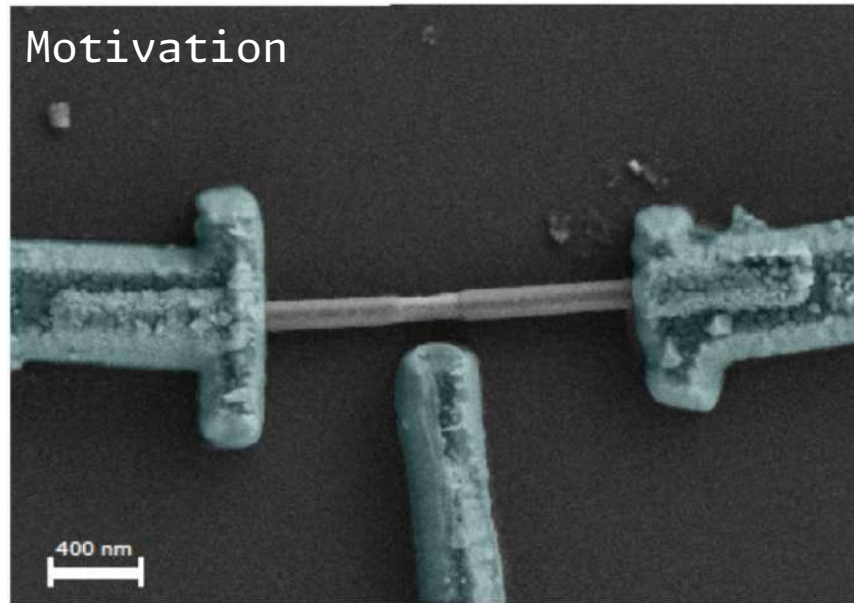
Quantica.jl

```
g = hN |> attach(gS[cells = (-1)], coupling1;) |> attach(gS'[cells = (1)], coupling2) |>  
greenfunction()
```

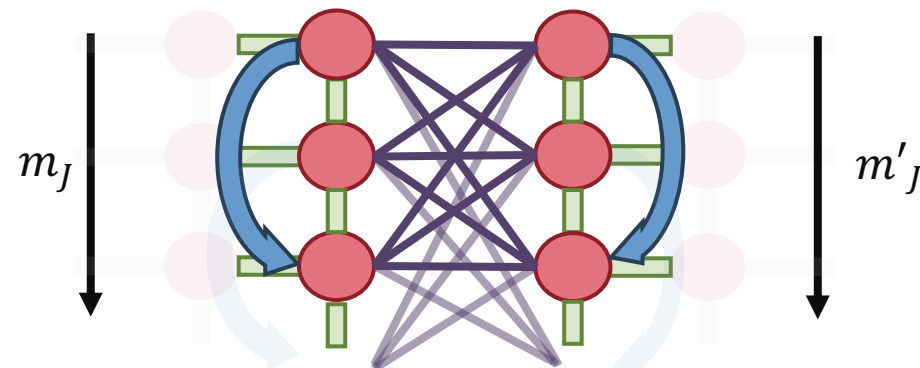
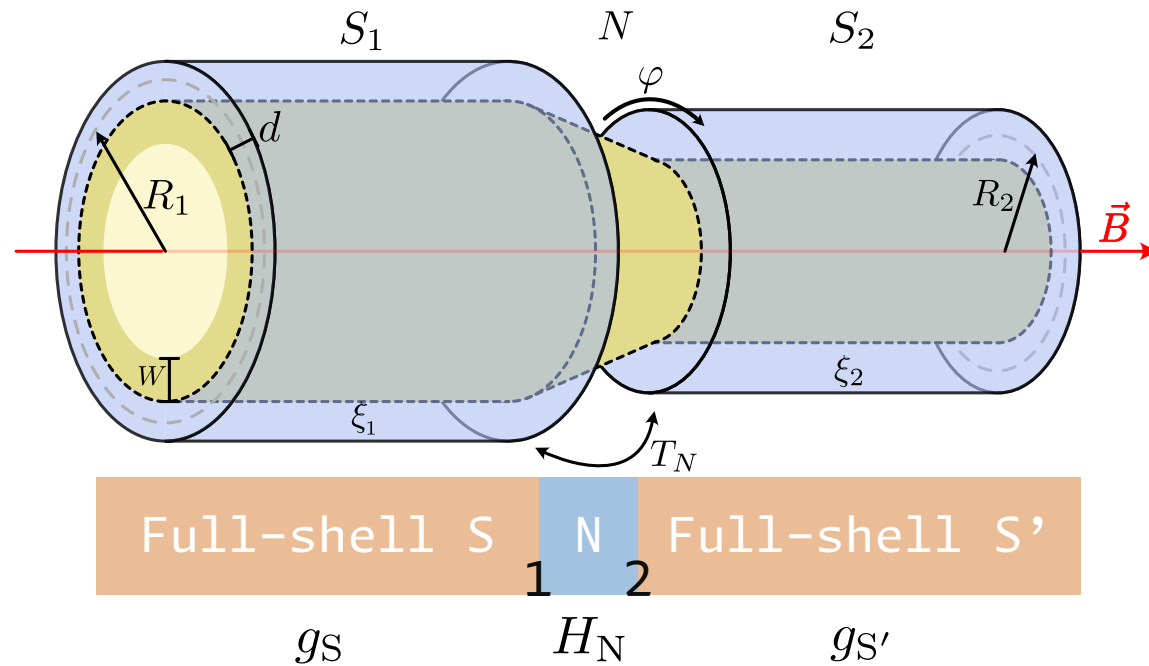
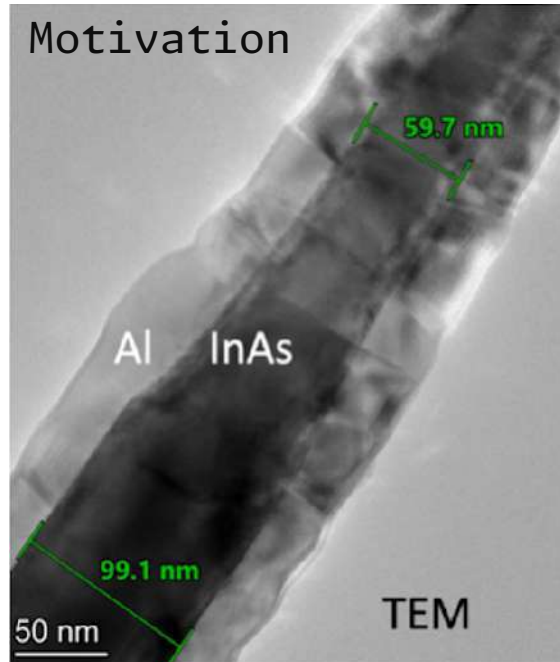
```
J = josephson(g[1], ipath)
```

```
J(kBT; params...)
```

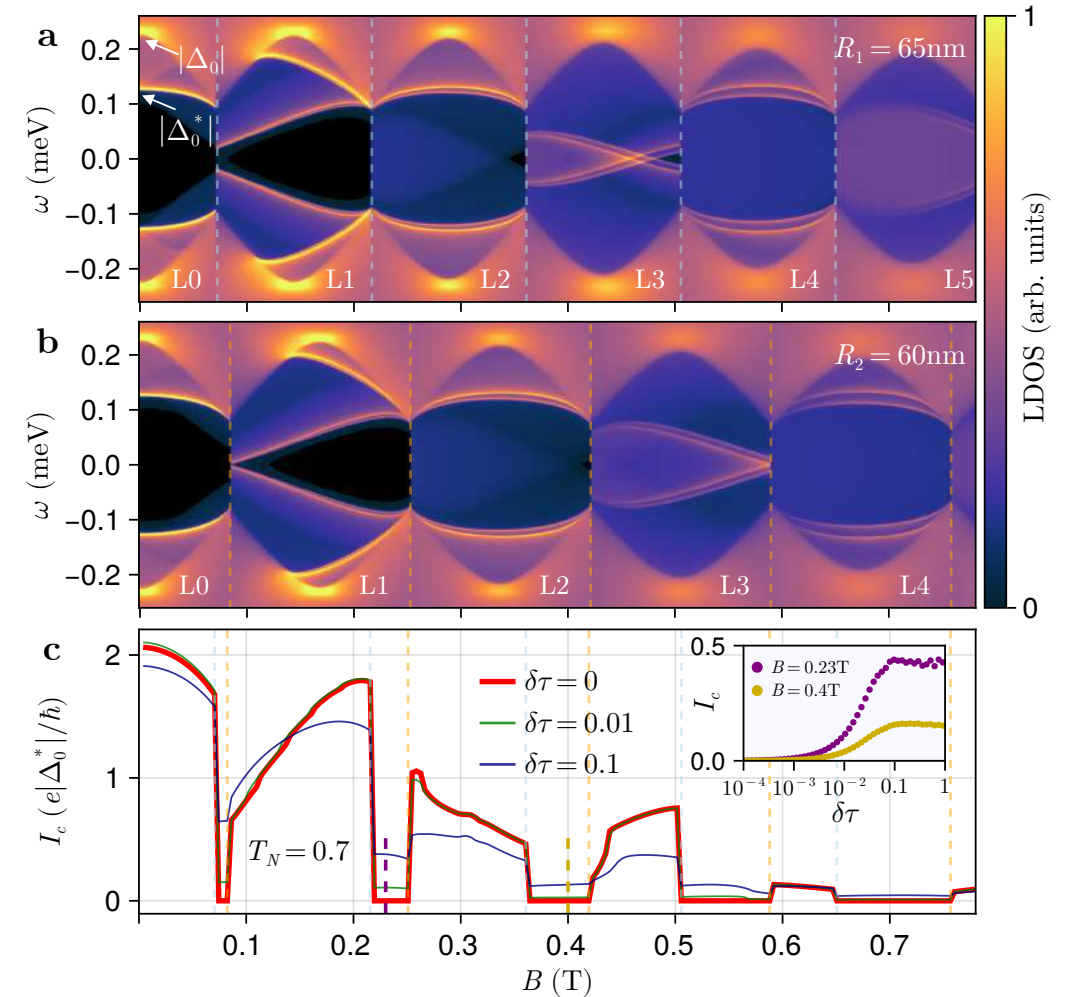
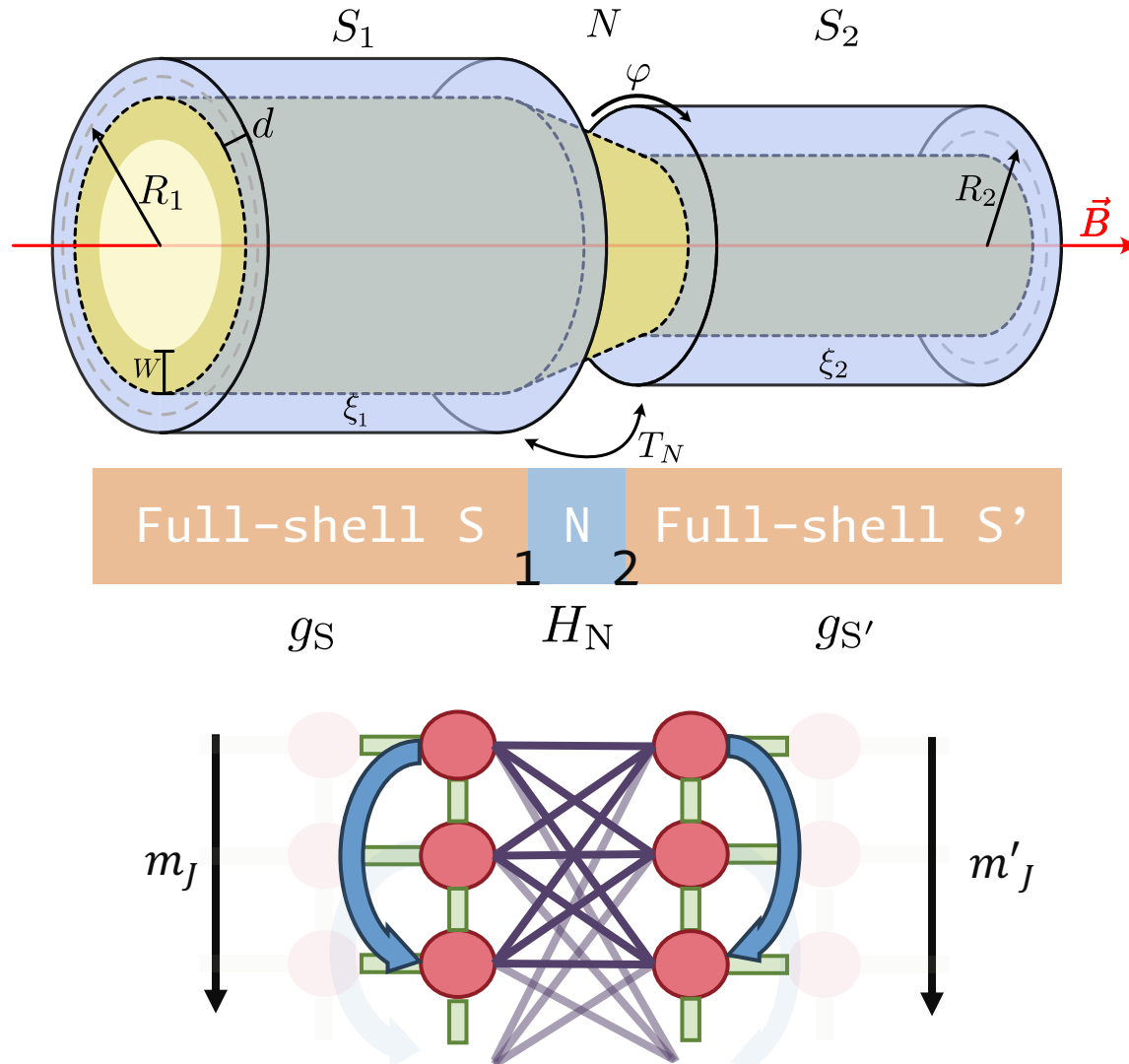

Building other setups: Josephson Junction



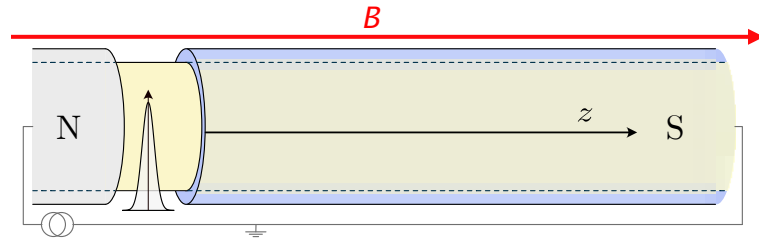
Building other setups: Fluxoid Valve



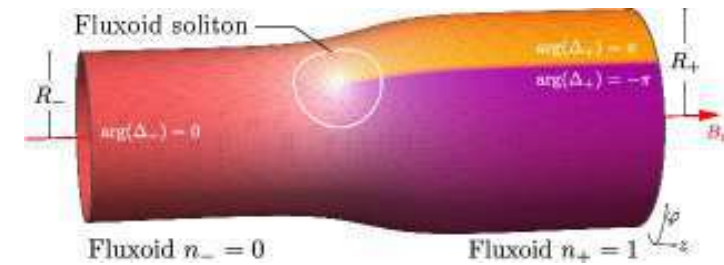
Building other setups: Fluxoid Valve



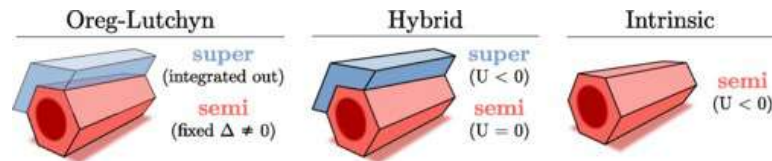
Quantica: simple and versatile



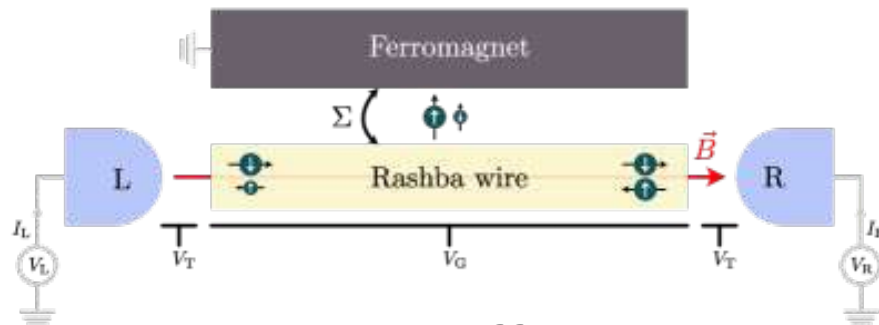
Full-Shell Project: CP thesis



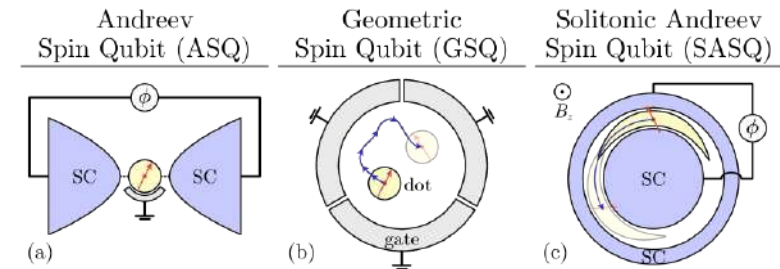
Tapered SC wires: T. Kokkeler *et al.*



Intrinsic Majorana wires: F. Lobo *et al*



Non-Hermitian Skin Effect: CP with K. Flensberg (*et al*)



Solitonic Andreev Spin Qubit: P. San-Jose and E. Prada



 carlosp24.github.io

 carlos.paya@csic.es

 @cPaya_phys

[Code for this tutorial](#)



Realistic Modeling of 1D Hybrid Superconductors

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